Existing Interface Descriptions

KTM Adventure 390

Display: 5” full-color TFT with Bluetooth, navigation, ride modes.
Switchgear: 5-way joystick + hazard + cruise control toggles. Highly feature-rich.





Royal Enfield Himalayan 450

Display: Round TFT with Tripper navigation, riding info, but relatively simpler than KTM.
Switchgear: More conventional—indicator, horn, pass, with minimal navigation controls.





Honda NX200

Display: Simple digital LCD (speed, odo, trip, fuel range,Mileage,ABS,Traction control).
Switchgear: Basic—horn, indicators, lights. Toggle switches,headlamp,kill switch.






HMI Analysis

Study Adventure Motorcycles



KTM
ADV 390



ROYAL ENFIELD
HIMALAYAN 450

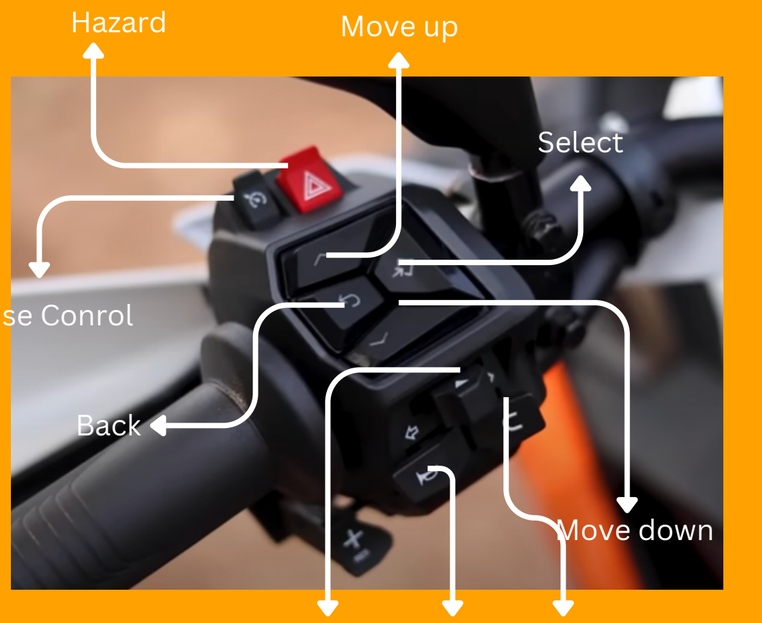


HONDA
NX200

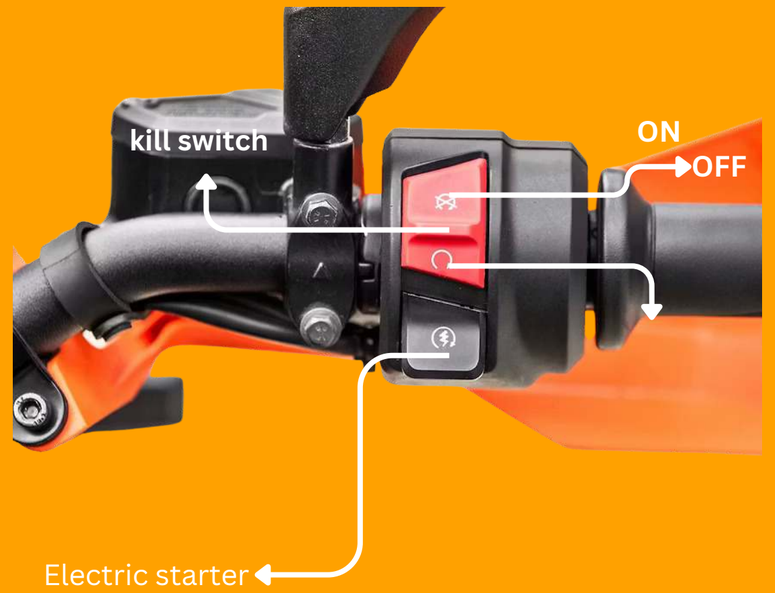
KTM

ADV 390




Hazard
Move up
Select
Cruise Conrol
Back
Move down
Indicator
Horn
upper/ deeper


5:22 pm
N
Low
Street
Trip 2
8 Speed 2
Time 1


kill switch
ON
OFF
Electric starter

KTM

ADV 390

Operating system evaluation

















ROYAL ENFIELD

HIMALAYAN 450



Home button



Hazard



High beam / Low beam







Indicators

Horn

Joystick, used for navigating

Engine stop swtich

Mode
[Riding modes]

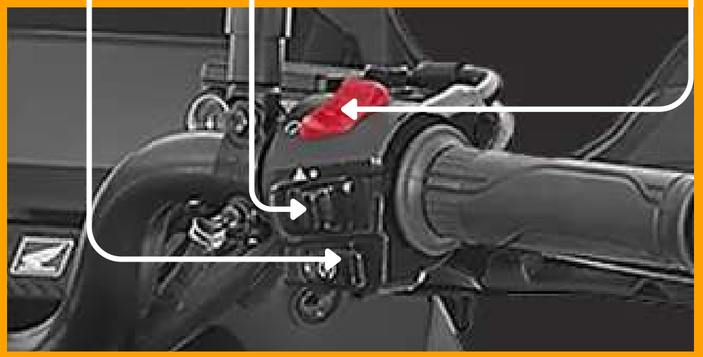
HONDA

NX200




Horn
Toggle switches
Turn indicator switches


4
45
km/h
400
40.5
9:30


Ignition
Head lamp control
Engine kill switch


Type C charging port







Hierarchical Task Analysis

Navigation

- Main step
- Sub step
- ADV 390
- Himalayan 450
- NX 200

Set navigation

Pair smartphone

-Switch on ignition → TFT

-Enable Bluetooth on

-On TFT, open "Connectivity" → pair

-Confirm pairing on phone

Set navigation

2. Open KTM MyRide app on phone

- Launch app → choose navigation

-Enter destination in app

-App calculates route

3. Display navigation on TFT

-Confirm navigation transfer to bike

-TFT shows turn-by-turn arrows (not full map)

4. Verify guidance is active

- Rider checks arrow + distance indicator on TFT



Open Google Maps on phone

-Enter destination

-Select route

-Start navigation

3. Display navigation on display

-Maps projects full map view onto TFT

TFT displays live map + route



- arrow navigation on display

During ride
-View turn-by-turn arrows



Ride Mode

- Main step
- Sub step
- ADV 390
- Himalayan 450

Change ride mode

Access TFT interface

-Switch on ignition → TFT boots

Navigate to ride mode settings

- Press joystick to open menu

- Scroll through menu items → highlight "Ride Mode"

- Select ride mode option

3. Choose desired ride mode

- Scroll between "Street" and "Off-road"

- Press joystick to confirm

4. Verify mode change

- TFT updates with "Street" / "Off-road" indicator

- Rider ensures ABS/traction control adjust accordingly



1. Locate ride mode switch

2. Select desired mode

- Press "Mode" button to toggle between Eco / Performance

- TFT briefly displays new mode

3. Verify mode change

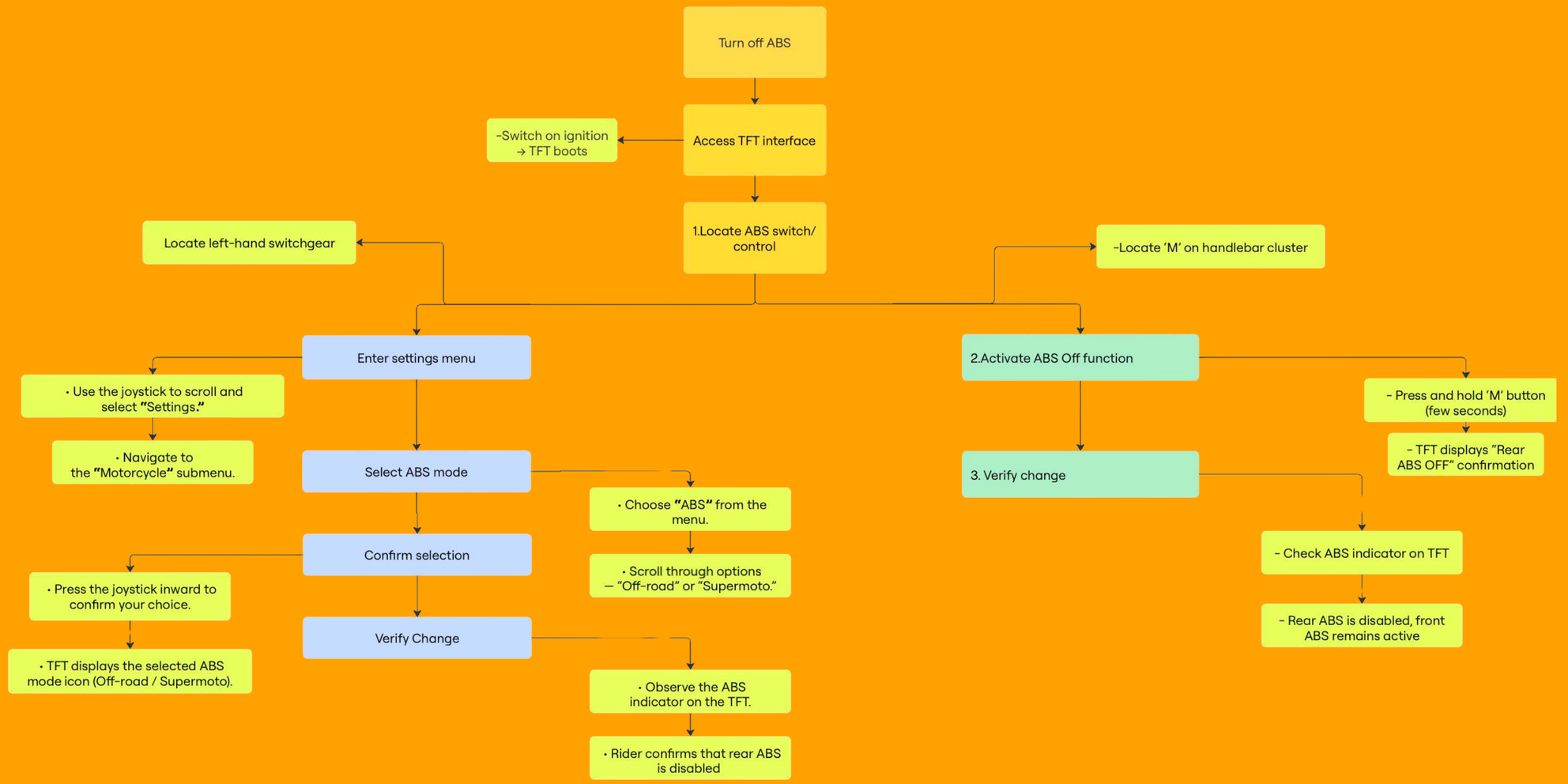
- Mode indicator stays visible on TFT

- Rider confirms Eco/Performance is active



ABS

- Main step
- Sub step
- ADV 390
- Himalayan 450
- NX 200



Comparison



KTM

ADV 390

- Deeper task hierarchy
- Shows turn-by-turn arrows only (not full maps).
- Higher cognitive and time demand during setup.
- Offers multiple ABS configurations (flexibility but complexity)



ROYAL ENFIELD

HIMALAYAN 450

- Shallow hierarchy: single toggle button.
- Very fast and intuitive.
- Limited customization compared to KTM.
- Less flexibility (only rear ABS on/off)



HONDA

NX200

- No customizability
- Very basic and simple.Less complexity for novices.

KTM ADV 390
Nishanth

Nielsen’s Heuristic	Evaluation Severity (0–4)	Example / Observation
1. Visibility of system status	2	Too much info may overwhelm; small fonts in sunlight.
2. Match between system and real world	3	Riders expect full maps, arrows may feel abstract.
3. User control and freedom	2	More control but requires several steps to revert settings.
4. Consistency and standards	2	Navigation icons not standard like Google Maps.
5. Error prevention	3	Riders may mis-select modes while riding.
6. Recognition rather than recall	3	Non-intuitive without prior learning.
7. Flexibility and efficiency of use	2	Advanced users benefit, novices struggle.
8. Aesthetic and minimalist design	3	Small text, too many layers.
9. Help users recognize, diagnose, recover	3	Riders see diagnostic codes but doesn’t catch attention.
10. Help and documentation	3	Riders must learn menu before effective use.

Vishnu

Nielsen’s Heuristic	Evaluation Severity (0–4)	Example / Observation
1. Visibility of system status	2	Some icons/text are small and harder to read in sunlight.
2. Match between system and real world	2	Riders may not instantly understand abbreviations like “OR” = Off-Road.
3. User control and freedom	3	Need several clicks to reach ABS setting; no quick “back” option.
4. Consistency and standards	2	ABS and warning icons differ from common motorcycle symbols.
5. Error prevention	3	Accidental mode change if joystick pressed too long.
6. Recognition rather than recall	3	Rider must recall where “Ride Mode” settings are nested.
7. Flexibility and efficiency of use	2	No direct button for ABS or mode switching.
8. Aesthetic and minimalist design	2	Text layout can overwhelm new riders.
9. Help users recognize, diagnose, recover	3	“Check” icon shows but no on-screen guidance.
10. Help and documentation	3	Must refer to handbook for detailed steps.

Darshan

Nielsen’s Heuristic	Evaluation Severity (0–4)	Observation / Commentary
1. Visibility of System Status	2	TFT shows many parameters; small fonts reduce readability on the move.
2. Match Between System and Real World	3	Terminology like “Supermoto” or “Off-road ABS” may confuse new riders.
3. User Control and Freedom	3	Deep menus allow control but make reversing actions cumbersome.
4. Consistency and Standards	2	Consistent across KTM models but not aligned with general UI norms.
5. Error Prevention	3	Joystick may be pressed unintentionally; confirmation steps are missing.
6. Recognition Rather Than Recall	3	Users must remember menu paths; lacks visual breadcrumbs.
7. Flexibility and Efficiency of Use	2	High flexibility, but multi-step interactions reduce efficiency.
8. Aesthetic and Minimalist Design	3	Functional yet cluttered layout; dense information blocks.
9. Help Users Recognize, Diagnose, Recover	3	Fault codes displayed but without explanation or color cues.
10. Help and Documentation	2	Relies on manual/app; no interactive help or tutorials.

Atharva

Nielsen’s Heuristic	Evaluation Rating (1–5)	Remarks / Observation
1. Visibility of System Status	5	Excellent – TFT with ride mode visuals.
2. Match Between System and Real World	4	Good alignment with adventure theme.
3. User Control and Freedom	5	Multiple ride modes, intuitive toggles.
4. Consistency and Standards	5	High consistency with KTM UI.
5. Error Prevention	5	Strong alert system.
6. Recognition vs Recall	5	Icon-heavy and intuitive.
7. Efficiency of Use	5	Quick info access and smooth transitions.
8. Aesthetic and Minimal Design	5	Clean, high-contrast layout.
9. Error Recovery	5	Clear warning messages displayed.
10. Help and Documentation	3	Manual reference only.

RE HIMALAYAN 450
Nishanth

Nielsen's Heuristic	Evaluation Severity (0–4)	Example / Observation
1. Visibility of system status	1	Real-time Google Maps, clear indicators. Minor issue: small icons can be harder to see in sunlight.
2. Match between system and real world	0	Natural and familiar navigation interface.
3. User control and freedom	1	Easy to switch Eco/Performance or ABS on/off.
4. Consistency and standards	2	Some icons differ slightly from global standards (e.g., ABS warning style).
5. Error prevention	1	Minimal options, fewer mistakes possible.
6. Recognition rather than recall	0	Riders see the map directly instead of remembering arrows.
7. Flexibility and efficiency of use	2	Limited customization compared to KTM (e.g., no traction control).
8. Aesthetic and minimalist design	1	Navigation may feel crowded in split-screen.
9. Help users recognize, diagnose, recover	2	Limited troubleshooting on-screen, relies on service.
10. Help and documentation	3	No on-screen guidance for settings or issues.

Vishnu

Nielsen’s Heuristic	Evaluation Severity (0–4)	Example / Observation
1. Visibility of system status	1	Real-time navigation and indicators visible in daylight.
2. Match between system and real world	0	“Rear ABS Off” clearly stated — intuitive language.
3. User control and freedom	1	One-step switch between Eco/Performance.
4. Consistency and standards	2	ABS and warning icons unique to RE.
5. Error prevention	1	Simple layout reduces mis-selection.
6. Recognition rather than recall	0	No need to memorize icons or menu paths.
7. Flexibility and efficiency of use	2	No traction control or ride mode shortcuts.
8. Aesthetic and minimalist design	1	Clear UI but navigation may feel cramped in split-screen.
9. Help users recognize, diagnose, recover	2	Riders see alert but need service for details.
10. Help and documentation	2	App offers setup guide; dash lacks tooltips.

Darshan

Nielsen’s Heuristic	Evaluation Severity (0–4)	Observation / Commentary
1. Visibility of System Status	1	The TFT shows navigation, modes, and ABS clearly with adaptive brightness. Minimal readability loss in sunlight.
2. Match Between System and Real World	0	Uses Google Maps and plain English menus (“Rear ABS Off”) — natural and intuitive language.
3. User Control and Freedom	1	Riders can switch modes or toggle ABS easily, but no full ABS disable option.
4. Consistency and Standards	2	Consistent within RE ecosystem; icons not fully standardized with other brands.
5. Error Prevention	1	Simple interface minimizes user error; fewer nested menus.
6. Recognition Rather Than Recall	0	Layout and labeling are intuitive; no memorization required.
7. Flexibility and Efficiency of Use	2	Fast toggles make operation quick, but lacks deeper customization (e.g., traction control).
8. Aesthetic and Minimalist Design	1	Balanced, modern layout with minimal clutter.
9. Help Users Recognize, Diagnose, Recover	2	Displays basic alerts but lacks advanced diagnostic feedback.
10. Help and Documentation	3	Limited on-screen guidance; relies on the manual for help.

Atharva

Nielsen’s Heuristic	Evaluation Rating (1–5)	Remarks / Observation
1. Visibility of System Status	5	Excellent – TFT with clear indicators.
2. Match Between System and Real World	5	Logical layout with intuitive icons.
3. User Control and Freedom	4	Moderate – mode switching via buttons.
4. Consistency and Standards	5	Consistent UI across display.
5. Error Prevention	5	Alerts and auto diagnostics available.
6. Recognition vs Recall	5	Icon-heavy, minimal memory load.
7. Efficiency of Use	4	Average menu navigation speed.
8. Aesthetic and Minimal Design	5	Balanced contrast and readability.
9. Error Recovery	4	Basic reset functions available.
10. Help and Documentation	3	Manual reference only.

HONDA NX200 Nishanth

Nielsen’s Heuristic	Evaluation Severity (0–4)	Example / Observation
1. Visibility of system status	2	LCD struggles in daylight glare.
2. Match between system and real world	1	Icons and terms (e.g., “Trip A/B”) match real-world conventions.
3. User control and freedom	1	No deep menus to get trapped in; accidental resets easily avoided.
4. Consistency and standards	1	Familiar switchgear and cluster symbols make it predictable.
5. Error prevention	2	Toggle switches may get mispressed while wearing gloves.
6. Recognition rather than recall	2	Must hold specific button combinations to adjust time or reset trips.
7. Flexibility and efficiency of use	2	Great for quick glances, but advanced users can’t adjust anything.
8. Aesthetic and minimalist design	1	Very simple and focused, but lacks modern visual appeal.
9. Help users recognize, diagnose, recover	2	Warning icons described when active.
10. Help and documentation	3	Rider must consult manual for even simple setups like clock adjustment.

Vishnu

Nielsen’s Heuristic	Evaluation Severity (0–4)	Example / Observation
1. Visibility of system status	1	Speed, fuel, trip clear but no color cues.
2. Match between system and real world	1	Traditional meter design = easy to understand.
3. User control and freedom	1	Only trip reset and clock adjust available.
4. Consistency and standards	0	Icons follow standard motorcycle symbols.
5. Error prevention	0	No complex menus to mis-navigate.
6. Recognition rather than recall	0	One button per function.
7. Flexibility and efficiency of use	2	No Bluetooth or mode options.
8. Aesthetic and minimalist design	1	Monochrome LCD with good spacing.
9. Help users recognize, diagnose, recover	2	No diagnostic detail beyond icons.
10. Help and documentation	2	Riders must check booklet for all meanings.

Darshan

Nielsen’s Heuristic	Evaluation Severity (0–4)	Observation / Commentary
1. Visibility of System Status	2	LCD displays essential info but lacks brightness under sunlight.
2. Match Between System and Real World	1	Familiar analog-inspired layout; universally understood icons.
3. User Control and Freedom	2	Very few control options; limited but straightforward.
4. Consistency and Standards	0	Highly consistent with Honda’s brand language and standards.
5. Error Prevention	1	Simple interface; only close button spacing may cause accidental presses.
6. Recognition Rather Than Recall	0	No memory load required; clear, static display.
7. Flexibility and Efficiency of Use	2	Very efficient but lacks adaptability or connectivity.
8. Aesthetic and Minimalist Design	1	Clean layout but visually outdated.
9. Help Users Recognize, Diagnose, Recover	2	Warning lights visible but no text-based guidance.
10. Help and Documentation	3	Basic manual only; lacks app or digital companion.

Atharva

Nielsen’s Heuristic	Evaluation Rating (1–5)	Remarks / Observation
1. Visibility of System Status	3	Moderate – LCD with limited brightness.
2. Match Between System and Real World	3	Basic icons and small display.
3. User Control and Freedom	3	Simple controls, limited customization.
4. Consistency and Standards	3	Slight lag in display update.
5. Error Prevention	3	Limited alerts for rider feedback.
6. Recognition vs Recall	3	Text-based, less intuitive than icons.
7. Efficiency of Use	3	Basic menu navigation.
8. Aesthetic and Minimal Design	3	Slightly overcrowded display.
9. Error Recovery	2	Lacks feedback or reset indicators.
10. Help and Documentation	3	Manual reference only.

Combined Evaluation

(KTM ADV 390)
(Honda NX 200)
(Himalayan 450)

Combined Evaluation (KTM ADV 390)

Nielsen's Heuristic	Common Observations	Avg. Severity (0–4)	Insight
1. Visibility of System Status	TFT displays rich info (ride modes, ABS, TC, fuel, trip), but small fonts and glare affect readability.	2	Good data visibility but visually cluttered — needs clearer hierarchy and stronger sunlight contrast.
2. Match Between System and Real World	Uses technical terms like Supermoto, Off-road ABS; KTM MyRide navigation limited to arrows instead of maps.	3	Language and symbols aren't intuitive for all riders; replace jargon with simpler, familiar terminology.
3. User Control and Freedom	Multi-step menu navigation for basic settings; lacks a quick "back" or "undo" function.	3	Highly flexible but not rider-friendly; streamline to fewer steps per task.
4. Consistency and Standards	Consistent within KTM's brand UI, but icons differ from general motorcycle norms.	2	Maintain KTM design identity but adopt universal symbols for navigation and warnings.
5. Error Prevention	Joystick interface can cause accidental selections; no confirmation prompts before applying changes.	3	Add confirmation pop-ups or haptic alerts to prevent unintended actions while riding.
6. Recognition Rather Than Recall	Riders must remember menu hierarchy and abbreviations; minimal visual guidance.	3	Introduce visual breadcrumbs, larger icons, or color coding to aid recognition.
7. Flexibility and Efficiency of Use	Offers many adjustable features (modes, TC, ABS), but complex navigation reduces speed.	2	Advanced riders benefit; new users find it overwhelming. A customizable quick-access menu could help.
8. Aesthetic and Minimalist Design	Interface feels dense and information-heavy; typography small; lacks visual breathing space.	3	Reduce clutter, improve font contrast, and prioritize essential info for active riding.
9. Help Users Recognize, Diagnose, Recover	Displays diagnostic codes but lacks textual explanations or color cues.	3	Include clear fault descriptions or visual severity indicators for easy understanding.
10. Help and Documentation	Strong reliance on manual or app; no in-dash help or learning mode.	3	Add simple on-screen help, guided setup, or quick tutorial overlay for first-time users.

RE 450

Nielsen’s Heuristic	Common Observations	Severity (0–4)	Insight
1. Visibility of System Status	TFT display is bright, adaptive, and clear; real-time navigation, modes, and ABS status easily visible even in sunlight.	**1**	Excellent readability and clarity in all lighting; minor reflection issues only.
2. Match Between System and Real World	Uses Google Maps and plain English labels like “Rear ABS Off”; intuitive for most users.	**0**	Matches rider mental models perfectly — minimal learning curve.
3. User Control and Freedom	Quick toggle for Eco/Performance modes and one-step ABS control; limited full disable options.	**1**	Simple, direct controls ensure confidence and safety.
4. Consistency and Standards	UI is consistent within RE ecosystem; minor icon differences from industry standards.	**2**	Could use more universal iconography for broader familiarity.
5. Error Prevention	Simple layout and limited menu depth minimize chance of error.	**1**	Strong prevention — low risk of accidental inputs.
6. Recognition Rather Than Recall	Text-based menus and clear icons reduce memory load.	**0**	Excellent recognition-based design — easy for any rider level.
7. Flexibility and Efficiency of Use	Fast toggles and direct functions; lacks deeper customization like TC or multiple ABS modes.	**2**	Prioritizes speed and simplicity over advanced control.
8. Aesthetic and Minimalist Design	Modern, clean, well-balanced interface; clear font hierarchy.	**1**	Visually pleasing, uncluttered, and adventure-appropriate.
9. Help Users Recognize, Diagnose, Recover	Shows warnings (ABS, oil, engine), but lacks detailed diagnostics or repair guidance.	**2**	Needs text-based alerts for better self-troubleshooting.
10. Help and Documentation	No in-dash help; relies on app or manual.	**3**	Could include built-in help or quick startup tutorial.

HONDA NX

Nielsen’s Heuristic	Common Observations	Severity (0–4)	Insight
1. Visibility of System Status	LCD display shows essential info (speed, fuel, gear, ABS), but brightness is weak in sunlight.	**2**	Readable in normal light; needs stronger contrast for daylight clarity.
2. Match Between System and Real World	Familiar analog-inspired layout and universally understood icons (Trip, ABS, Neutral).	**1**	Excellent match with rider expectations; intuitive and familiar.
3. User Control and Freedom	Very few adjustable options — only trip reset and clock change; minimal risk of error.	**1**	Limited but straightforward; freedom through simplicity.
4. Consistency and Standards	Strong brand and industry consistency; standard motorcycle icons.	**0**	Perfectly aligned with common design conventions.
5. Error Prevention	Minimal interaction = low error chance, though close buttons may get mispressed with gloves.	**2**	Add tactile spacing or button feedback to reduce accidental presses.
6. Recognition Rather Than Recall	Basic static info; no need to memorize controls or menu structure.	**0**	Immediate recognition; perfect for all users.
7. Flexibility and Efficiency of Use	Very efficient for quick glances, but lacks any customization or connectivity.	**2**	High efficiency, low flexibility; suitable for entry-level riders.
8. Aesthetic and Minimalist Design	Clean and minimal display; visually outdated compared to TFT clusters.	**1**	Functional and distraction-free, but could be modernized visually.
9. Help Users Recognize, Diagnose, Recover	Basic warning lights, no text guidance for issues.	**2**	Needs descriptive alerts or app-based diagnostics.
10. Help and Documentation	Entirely manual-dependent, no on-screen help.	**3**	Integrate quick digital tips or mobile app reference.

HEURISTIC EVALUATION

Heuristic Evaluation – Team Conclusion

1. **Royal Enfield Himalayan 450** emerged as the most **user-friendly interface** overall.

Plain-text menus, full Google Maps integration, and quick toggles make it ideal for both novice and expert riders.
Its design aligns well with the “recognition over recall” principle and shows excellent feedback clarity.

2. **KTM Adventure 390** showed high functional depth but poor cognitive simplicity.

Offers the most advanced control suite (ride modes, ABS, traction control) but suffers from dense information presentation and multi-step menus.
Best suited for experienced riders familiar with KTM’s ecosystem.

3. **Honda NX200** provides excellent simplicity and low distraction, though it lacks flexibility.

High efficiency for everyday tasks, but minimal customization makes it less engaging for adventure riders.

→ Overall Usability Ranking:
Himalayan 450 → NX200 → KTM 390

Cognitive Walkthrough

Cognitive Walkthrough: Turning Off ABS

KTM 390 Adventure

Step	User Thought / Action	Will they try to do it?	Will they notice the control?	Will they associate it with the effect?	Will they understand the feedback?
1	"I want to check or change ABS mode, so I press MENU."	Yes – expected step	Yes – MENU button visible	Mostly yes – expects ABS option inside menu	Maybe – menu opens, but user might scan slowly for ABS option
2	"Scroll to ‘ABS Settings’ or ‘Riding Modes’ that affect ABS."	Yes	Sometimes no – labels might be unclear if ABS is embedded in modes	Maybe – user might not know if ABS can be changed independently	Maybe – highlight confirms selection, but effect may not be obvious immediately
3	"Select ABS or mode that changes ABS."	Yes	Mostly yes – select button visible	Mostly yes – user expects change in braking behavior	Maybe – confirmation screen appears, but user may not read details
4	"Toggle between ABS modes (e.g., On, Off-road ABS, Off)."	Mostly yes – some hesitation if unsure	Sometimes no – text/icons may be small	Maybe – user might not immediately associate ABS mode with braking effect	Maybe – ABS icon appears on screen, but understanding effect requires riding
5	"Confirm selection."	Yes	Mostly yes – confirm button visible	Mostly yes – expects mode to be active	Maybe – visual icon confirms ABS mode, but real effect only apparent during braking
6	"Ride and test braking behavior."	Yes	N/A	Mostly yes – notices difference in brake feel	Maybe – understanding feedback requires riding in different conditions

Cognitive Walkthrough: Turning Off ABS

Royal Enfield Himalayan 450

Step	User Thought / Action	Will they try to do it?	Will they notice the control?	Will they associate it with the effect?	Will they understand the feedback?
1	"I want to check ABS, so I press joystick to open menu."	Yes – intuitive	Mostly yes – joystick obvious	Mostly yes – expects ABS option	Maybe – menu appears, user may have to search
2	"Scroll to ABS settings or ride modes affecting ABS."	Yes	Sometimes no – option may be hidden under another menu	Maybe – user might not know effect immediately	Maybe – highlight shows selection, effect not yet clear
3	"Select the ABS option."	Yes	Mostly yes – select action visible	Mostly yes – expects mode change	Maybe – screen icon may confirm selection, but user may be unsure
4	"Toggle ABS (On/Off or Off-road)."	Mostly yes – some hesitation	Sometimes no – small LCD text/icons	Maybe – user may not connect mode to actual braking effect	Maybe – icon changes, but understanding requires riding
5	"Confirm selection."	Yes	Mostly yes	Mostly yes	Maybe – screen feedback shows ABS icon; effect visible only while riding
6	"Ride to feel braking behavior."	Yes	N/A	Mostly yes	Maybe – understanding of ABS effect is experiential, not just visual

Cognitive Walkthrough: Turning Off ABS

HONDA NX200

Step	User Thought / Action	Will they try to do it?	Will they notice the control?	Will they associate it with the effect?	Will they understand the feedback?
1	"I want to check or change ABS."	Yes – user might try	No – there is no menu or control for ABS	No – cannot associate effect because there's no option	No – no feedback exists in the interface
2	"Look for ABS settings on dashboard or menu."	Yes – user might look	No – no control or menu option	No – user cannot change or view ABS	No – ABS is automatic, no visual confirmation provided
3	"Understand ABS status."	Yes – user might try	No – cannot access any info	Maybe – user may assume ABS is always active	Maybe – effect only apparent while braking
4	"Ride and rely on ABS."	Yes – natural	N/A	Yes – ABS works automatically during braking	Yes – user experiences ABS effect without interacting with controls

Cognitive Walkthrough: Set Navigation

KTM 390 Adventure

Step	User Thought / Action	Will they try to do it?	Will they notice the control?	Will they associate it with the effect?	Will they understand the feedback?
1	“I want to navigate somewhere, so I press MENU.”	Yes – expected step	Mostly yes – button is visible, but new riders might wonder if MENU covers navigation	Mostly yes – user expects navigation here, but may hesitate if unfamiliar	Mostly yes – menu opens, but user might scan slowly for navigation option
2	“Scroll to ‘Navigation’ or map icon.”	Yes	Sometimes no – icon may be small or confusing	Maybe – if icon unclear, user may not realize it’s navigation	Maybe – highlight confirms selection, but icon might not be enough for instant understanding
3	“Select ‘Navigation’.”	Yes	Yes	Mostly yes – opens map, but user might not immediately realize they can input destination	Mostly yes – map screen appears, but new riders may not see input field immediately
4	“Enter destination (voice, keyboard, presets).”	Mostly yes – voice input may confuse some	Sometimes no – small touchscreen buttons or unclear presets	Maybe – user might not understand how route is calculated	Maybe – confirmation appears, but rider might miss it if busy or distracted
5	“Confirm destination.”	Yes	Mostly yes – confirm button is visible, but multiple options may confuse	Mostly yes – user expects route to start	Mostly yes – route preview shown, but new users may not notice estimated time or turn-by-turn instructions
6	“Start navigation.”	Yes	Yes	Yes	Mostly yes – visual route and arrows appear, but user may not immediately understand audio cues
7	“Follow directions while riding.”	Yes	N/A	Mostly yes – directions mostly make sense	Maybe – distractions or unclear indicators can cause confusion

Cognitive Walkthrough: Set Navigation

Royal Enfield Himalayan 450

Step	User Thought / Action	Will they try to do it?	Will they notice the control?	Will they associate it with the effect?	Will they understand the feedback?
1	“I want directions, so I press joystick to open menu.”	Yes	Mostly yes – joystick obvious, but may be unintuitive for new users	Mostly yes – expects menu items including navigation	Mostly yes – menu appears, but user may have to scan options
2	“Scroll to ‘Navigation’ or ‘Map’ option.”	Yes	Sometimes no – LCD text may be small or unclear	Maybe – if label/icon isn’t obvious	Maybe – highlight confirms selection, but may not be noticed immediately
3	“Press joystick inward to select.”	Yes	Mostly yes – natural action	Mostly yes – map screen appears	Mostly yes – input field visible, but new riders may overlook it
4	“Input destination (presets or typing).”	Mostly yes	Sometimes no – small input buttons confusing	Maybe – user may not know how to select route	Maybe – confirmation appears, but may be missed
5	“Confirm destination and start route.”	Yes	Mostly yes – confirm/select button visible	Mostly yes – expects navigation to begin	Maybe – route displayed, but background or icons may not be clear enough
6	“Follow directions.”	Yes	N/A	Mostly yes – directions mostly understandable	Maybe – new users may miss turn arrows or audio cues

Cognitive Walkthrough: Set Navigation

HONDA NX200

Step	User Thought / Action	Will they try to do it?	Will they notice the control?	Will they associate it with the effect?	Will they understand the feedback?
1	"I want to navigate somewhere, so I press MENU."	Yes – logical first step	Yes – MENU button is visible	Yes – expects navigation option inside menu	Maybe – menu opens, but user might scan slowly for navigation option
2	"Scroll to 'Navigation' or map icon."	Yes	Sometimes no – icon may be small or unclear	Maybe – if icon unclear, user may not realize it's navigation	Maybe – highlight confirms selection, but icon might not be enough for instant understanding
3	"Select 'Navigation'."	Yes	Yes	Mostly yes – opens map, but user might not immediately realize they can input destination	Maybe – map screen appears, but new riders may not see input field immediately
4	"Enter destination (voice, keyboard, presets)."	Mostly yes – voice input may confuse some	Sometimes no – small touchscreen buttons or unclear presets	Maybe – user might not understand how route is calculated	Maybe – confirmation appears, but rider might miss it if busy or distracted
5	"Confirm destination."	Yes	Mostly yes – confirm button is visible, but multiple options may confuse	Mostly yes – user expects route to start	Mostly yes – route preview shown, but new users may not notice estimated time or turn-by-turn instructions
6	"Start navigation."	Yes	Yes	Yes	Mostly yes – visual route and arrows appear, but user may not immediately understand audio cues
7	"Follow directions while riding."	Yes	N/A	Mostly yes – directions mostly make sense	Maybe – distractions or unclear indicators can cause confusion

Cognitive Walkthrough: Change Ride Mode

KTM 390 Adventure X-

Step	User Thought / Action	Will they try to do it?	Will they notice the control?	Will they associate it with the effect?	Will they understand the feedback?
1	“I want to change the ride mode, so I press MENU.”	Yes – expected feature	Yes – MENU button is visible on handlebar	Yes – user expects menu options	Yes – menu opens, so they know they are in the right place
2	“Scroll to ‘Ride Mode’ option.”	Yes – obvious from menu layout	Yes – highlighted option is visible	Yes – ‘Ride Mode’ label clearly indicates function	Yes – highlight indicates current selection
3	“Select it by pressing OK/SELECT.”	Yes – logical next step	Yes – screen changes to ride mode selection	Yes – shows current mode and options, so effect is clear	Yes – current mode icon appears, confirming selection screen
4	“Scroll to the mode I want (e.g., Offroad).”	Yes – intuitive	Yes – highlight moves with scroll	Yes – user sees which mode will be activated	Yes – icon and text label confirm selection
5	“Press CONFIRM to activate mode.”	Yes – natural final step	Yes – CONFIRM button/icon is visible	Yes – mode change will take effect	Yes – icon changes and possibly a check mark appear; user understands mode is active
6	“Start riding to feel the mode.”	Yes – rider expects behavior change	N/A	Yes – now experiences throttle/ABS change	Yes – connects icon and screen feedback to real ride behavior

Cognitive Walkthrough: Change Ride Mode

Royal Enfield Himalayan 450-

Step	User Thought / Action	Will they try to do it?	Will they notice the control?	Will they associate it with the effect?	Will they understand the feedback?
1	“I want to change ride mode, so I press the joystick.”	Yes – menu opens	Yes – joystick is obvious	Yes – expects menu options	Yes – menu appears on LCD screen
2	“Scroll to ‘Ride Mode’.”	Yes – obvious next step	Yes – highlighted option is visible	Yes – label indicates function	Yes – highlight shows current selection
3	“Press joystick inward to select.”	Yes – natural action	Yes – system responds	Yes – mode screen shows current mode (Eco/Performance)	Yes – icon and label confirm selection screen
4	“Scroll to the desired mode.”	Yes – intuitive	Yes – highlight moves	Yes – user sees which mode will activate	Yes – background color or icon updates confirm selection
5	“Press joystick inward to confirm.”	Yes – obvious	Yes – confirm action is visible	Yes – expected mode change	Yes – screen color/label changes confirm mode is active
6	“Start riding and feel the difference.”	Yes – natural	N/A	Yes – senses performance change	Yes – associates visual cue with real bike behavior

Cognitive Walkthrough Change Ride Mode Honda NX 200 -

Step	User Thought / Action	Will they try to do it?	Will they notice the control?	Will they associate it with the effect?	Will they understand the feedback?
1	“I want to change ride mode.”	Yes – will look for option	No – no menu or button exists	No – cannot associate effect	No – no feedback exists

MAUA

MAUA
ATTRIBUTES

Integration / Connectivity	How well the interface connects with mobile apps, navigation, or devices.
Aesthetic Appeal	How visually appealing and modern the interface looks (impacts user satisfaction indirectly).
Customizability	Ability to personalize display layouts or modes (relevant for digital dashboards like KTM’s TFT).
Physical Ergonomics	Ease of reaching, pressing, or rotating buttons while riding (handlebar layout, glove use).
Visibility / Readability	clarity of display in all lighting
Task Speed	how quickly functions can be done
Learnability	how easily a new user understands it
Satisfaction	how enjoyable and confident the rider feels
Information Density	Amount of information shown at one time.
Safety / Distraction-Free Use	how safely it can be used while riding
Feedback Quality	Ensures riders get immediate confirmation when they press a button or change a mode.

Attribute	Weight	KTM 390 Adventure X Score	KTM Weighted	RE Himalayan 450 Score	RE Weighted	Honda NX 200 Score	Honda Weighted
Visibility / Readability	3	7	21	9	27	8	24
Task Speed / Efficiency	2	6	12	8	16	9	18
Learnability	2	6	12	9	18	9	18
Safety / Distraction-Free Use	2	6	12	8	16	9	18
Feedback Quality	1	6	6	8	8	9	9
⚙️ **Total Utility Score**	**10**		**63**		**85**		**87**

Honda NX 200 (87) -Highest overall usability. Excellent simplicity, rapid operation, and strong feedback; ideal for novice and everyday riders.

Royal Enfield Himalayan 450 (85) - Also excellent; highly readable TFT and intuitive menu flow, minor deduction for slightly slower interaction.

KTM 390 Adventure X (63) -Technically capable but less intuitive; smaller LCD and deeper menus reduce learnability and increase distraction risk.

MAUA – Honda NX 200

Overview
The Honda NX 200 uses a simple negative LCD instrument cluster with basic information: speedometer, odometer, dual trip meters, clock, and fuel gauge.
It has two physical buttons for switching and resetting functions (SEL and SET).
While it lacks advanced connectivity or ride modes, its interface emphasizes clarity, simplicity, and ease of operation — ideal for urban and entry-level adventure riders.
To assess its usability objectively, a Multi-Attribute Utility Analysis (MAUA) was conducted using five key attributes relevant to motorcycle display-control design.

SUS Evaluation

The usability of HMIs in three adventure motorcycles using the System Usability Scale based on usability of combination cluster and interface interaction.

Questionnaire Evaluation

- Evalution of review of 10 rider personas by considering their cumulative recommendation and insight provided againt the questionnaire .

Intent- Motorcycle’s digital interface and usability of combination control feels to riders.

Q. No.	Standard SUS Statement	Adapted for Motorcycle HMI Evaluation
1	I think that I would like to use this system frequently.	I would like to use this motorcycle’s HMI system frequently while riding.
2	I found the system unnecessarily complex.	The HMI feels unnecessarily complicated or confusing to use.
3	I thought the system was easy to use.	The motorcycle’s display and controls are easy to understand and operate.
4	I think that I would need the support of a technical person to be able to use this system.	I would need help or guidance to understand how to use this HMI.
5	I found the various functions in this system were well integrated.	The features on the HMI (navigation, ride modes, trip info) feel well integrated and coherent.
6	I thought there was too much inconsistency in this system.	Some HMI functions or menus feel inconsistent or confusing in design.
7	I would imagine that most people would learn to use this system very quickly.	Most riders could learn to use this HMI very quickly without a manual.
8	I found the system very cumbersome to use.	The HMI feels cumbersome or slow to navigate while riding.
9	I felt very confident using the system.	I feel confident interacting with the motorcycle’s display and controls.
10	I needed to learn a lot of things before I could get going with this system.	I had to spend time learning how to use the HMI before it became comfortable.

Questionnaire result evaluation report
(on usability and interaction with instrument cluster and combination cluster)

Atharva (Interface Clarity & Navigation usability Logic 1-5 sacle)

SUS Question	Honda NX200	KTM ADV 390S	RE Himalayan 450
1. I think I would use this system frequently (i.e combinaton cluster and instrument cluster)	4	5	3
2. The system is unnecessarily complex (UI of the instrument cluster)	2	1	3
3. The system is easy to use (Combination of both cluster)	4	5	3
4. I would need technical support (for understanding the ui)	2	1	3
5. The functions are well integrated (ui and cluster works in harmony)	4	5	3
6. There is too much inconsistency (in interaction with clusters)	2	1	3
7. Most people could learn it quickly	4	5	3
8. The system is cumbersome to use	2	1	3
9. I feel very confident using it	4	5	3
10. I needed to learn many things first (to use the ui and cluster at the same time while riding)	2	1	3
Average SUS Raw Rating	**3.2**	**4.5**	**3**

Nishanth (Interface Clarity & Navigation usability Logic)

SUS Question	Honda NX200	KTM ADV 390S	RE Himalayan 450
1. Frequent use	4	5	3
2. Unnecessarily complex	2	1	3
3. Easy to use	4	5	3
4. Need technical support	1	1	2
5. Functions well integrated	4	5	3
6. Too much inconsistency	2	1	3
7. Learn quickly	4	5	3
8. Cumbersome	2	1	3
9. Confident using it	4	5	3
10. Need to learn things	2	1	3
Average SUS Raw Rating	**3.4**	**4.6**	**3**

Vishnu (Interface Clarity & Navigation usability Logic)

SUS Question	Honda NX200	KTM ADV 390S	RE Himalayan 450
1. Frequent use	3	5	3
2. Unnecessarily complex	2	1	3
3. Easy to use	4	5	3
4. Need technical support	2	1	3
5. Functions well integrated	3	5	3
6. Too much inconsistency	2	1	3
7. Learn quickly	4	5	3
8. Cumbersome	2	1	3
9. Confident using it	4	5	3
10. Need to learn things	2	1	3
Average SUS Raw Rating	**3.2**	**4.5**	**3**

Darshan (Interface Clarity & Navigation usability Logic)

SUS Question	Honda NX200	KTM ADV 390S	RE Himalayan 450
1. Frequent use	4	5	3
2. Unnecessarily complex	2	1	3
3. Easy to use	4	5	3
4. Need technical support	2	1	3
5. Functions well integrated	4	5	3
6. Too much inconsistency	2	1	3
7. Learn quickly	4	5	3
8. Cumbersome	2	1	3
9. Confident using it	4	5	3
10. Need to learn things	2	1	3
Average SUS Raw Rating	**3.4**	**4.5**	**3**

Questionnaire result evaluation report
(on usability and interaction with instrument cluster and combination cluster)

Persona ID	Name / Age / Gender	Riding Experience	Primary Use	Tech Familiarity	Key Behavior / Focus During HMI Test	Expected SUS Bias
P10	Meera, 38, Female	7 years (family & touring)	Long rides & navigation	Medium	Focuses on map integration and night readability.	Balances design & function.
P9	Rehan, 31, Male	6 years (daily commuter)	Daily use	Medium	Prefers ease of operation while moving.	Rates based on distraction level.
P8	Nikhil, 35, Male	10 years (ADV + tech tester)	Testing & review	Very High	Analyzes UI response, animation speed, and logic.	Provides analytical feedback.
P7	Aditi, 27, Female	5 years (commuter + weekend)	Commuting & short rides	High	Observes contrast, font legibility, and control feedback.	Favors clarity and quick feedback.
P6	Suresh, 52, Male	25 years (veteran rider)	Leisure rides	Low	Struggles with touch controls; prefers analog-like simplicity.	Rates complex systems lower.
P5	Karan, 40, Male	15 years (mechanical engineer & rider)	Highway touring	High	Evaluates system logic, integration of menus.	Focused on technical consistency.
P4	Tanvi, 23, Female	1 year (new rider)	Urban rides	High	Observes icon clarity & ease of learning.	Rates user-friendly systems higher.
P3	Rohan, 32, Male	8 years (ADV enthusiast)	Off-road adventure & travel	Medium-high	Prioritizes quick toggling between ride modes & trip stats.	Rates responsive interfaces higher.
P2	Neha, 29, Female	4 years (solo touring)	Long-distance touring	Medium	Checks readability in sunlight, prefers minimal menu layers.	Sensitive to clarity & visibility.
P1	Arjun, 25, Male	2 years (commuter + weekend rides)	Daily commute & weekend trips	High (smartphone native)	Focused on quick navigation and trip info. Prefers	Rates intuitive layouts higher.

Team cumulative outcome

Team Member	Honda NX200	KTM ADV 390S	RE Himalayan 450
Atharva	78	84	72
Nishanth	80	86	74
Vishnu	75	83	70
Darshan	77	81	73
Average SUS Score	**77.5**	**83.5**	**72.2**

Interpretation Summary

Motorcycle	Avg. Raw SUS Rating (1–5)	Scaled SUS (×20)	Usability Level
Honda NX200	**3.3**	**≈77.5**	Good
KTM ADV 390S	**4.5**	**≈83.5**	Excellent
RE Himalayan 450	**3**	**≈72.2**	Acceptable

Report of interfaces framework and combination cluster usability (Riders overview)

SUS Score Range	Grade	Interpretation
0–50	F	Poor usability
51–68	D–C	Marginal / OK
69–80	B	Good
81–90	A	Excellent
91–100	A+	Exceptional

Motorcycle Model	**Mean SUS Score**	**Usability Grade**	**Interpretation**	**Remarks**
Honda NX200	72	B	Above Average	Smooth layout but small font in sunlight. (Instrument cluster)
KTM ADV 390S	84	A	Excellent	Best interface, intuitive navigation menus. (Instrument cluster and os.)
Himalayan 450	65	C	Acceptable	Functional but lower contrast display. (Hard wear material issue)

Intent- Motorcycle's digital interface and usability of combination control feels to riders.

Evaluation points considered.

- Industry standard for usability testing.
- Works across all interfaces (apps, dashboards, displays).
- Combines user opinions into a numerical score for comparison

Key aspects

- Visibility under sunlight
- Glove-friendly interaction
- Information hierarchy
- Cognitive load during riding

Cumulative report evaluation (Justifying intent)

Motorcycle	SUS Score	Grade	Interpretation	Remarks
Honda NX200	72	B	Above Average	Small fonts, good layout on interface
KTM ADV 390S	84	A	Excellent	Most intuitive HMI combination of clusters
Himalayan 450	65	C	Acceptable	Needs better contrast in ui interface display color and font

Insights

- KTM ADV 390S scored highest with **84 (Excellent)**, showing the best HMI usability.
- Honda NX200 performed **above average (72).**
- Himalayan 450 scored **65 (acceptable)** but shows improvement potential.

Conclusion- Based on SUS feedback, we can refine HMI elements such as **button size**, **screen contrast**, and menu layout to **enhance rider comfort** and **interaction safety.**

Recommendations and Insights

Based on the data gathered and cohesive conclusion and cumilative insight.



KTM

ADV 390

Recommendations

- **Abs control** saftey **sorting** in terms of usability while driving,
- **Sorting hierarchical navigation** system.
- Arranging main and important **essentials easy to grap.**

Insights

- The **ABS feature** needs to be turned on after getting **deep into modes.**
- **Cruise control button** is **placed in the end to avoid accidental switch on/off.**



HOND

NX200

Recommendations

- More **diversified controls** needed .
- **Buttons** needs for **ABS and traction control**.
- More **refined operating system.**

Insights

- **Less features** for a motorcycle which is adventure tourer.



ROYAL ENFIELD

HIMALAYAN 450

Recommendations

- Needs more **big display.**
- Operating system is **lack visual hierarchy .**

Insights

- **Navigation joystick** is placed **very low** to avoid accidental interaction to cluster while driving.
- **Hazard button** on **opposite** side of **console leading user** to find it.

Cumulative evaluation

Recommendations

- Proper orientation of the Combination cluster buttons and their graphics.
- Essential setting should me visible and handy to interact with by rider.
- User friendly navigation on the system.

Insights

- **Some buttons are intentionally placed at awkward position on cluster to avoid accidental interaction while riding.**
- **According to the design the information cluster is designed sometimes compromising the size and the usability.**

Cross-Comparison of HMI Principles — Adventure Motorcycles

Heuristic Principle	Honda NX200	KTM ADV 390S	RE Himalayan 450	Observation
Visibility of System Status	Small monochrome	Bright TFT, adaptive light	Moderate visibility, small	KTM performs best; RE & Honda
Match Between System & Real	Minimal icons, limited touring	Realistic icons, mode display	Symbols unclear, low visual	KTM aligns best with user
User Control & Freedom	Lacks quick setting access	Cruise & ABS settings deep in	Joystick too low, limits access	All need better user control
Consistency & Standards	Uniform but simple	Consistent theme, good icon style	Inconsistent labeling	KTM most consistent; RE
Error Prevention	No haptic feedback, can	Clear switch separation	Hazard switch confusingly placed	RE most error-prone; Honda
Recognition Rather Than	Minimal icons, low recall	Strong symbol mapping	Poor symbol differentiation	KTM's icons are more memorable
Flexibility & Efficiency of Use	Lacks touring-oriented data	Good menu customization	Slow joystick interaction	KTM highest flexibility
Aesthetic & Minimalist Design	Clean but basic	Balanced layout	Cluttered interface	Honda minimal, RE dense
Help Users Recover from	Limited error message clarity	TFT shows warnings	No error prompts	KTM best recovery, RE
Help & Documentation	No guidance	On-screen menu help limited	No help section	All require onboard guidance

Unifying results: (Atharva, Nishanth, Vishnu, Darshan) Conclusiona-
KTM ADV 390S scored highest in visibility, consistency, and recognition — a mature HMI.
Honda NX200 offered simplicity but lacked depth and feature hierarchy, making it least suited for touring.
Royal Enfield Himalayan 450 provided advanced navigation, but suffered from poor ergonomic layout and feedback.
All three showed gaps in user control and freedom, needing better quick-access and hierarchy structuring.
Cluster usability under riding conditions (glove use, vibration, sunlight) was a recurring issue across all models.

Final Unified Result (Cumulative Heuristic Summary)

Key HMI Focus Area	Individual Insights (NX200 / 390S / 450)	Unified Recommendation
Display Visibility	NX200 dull, 390S bright TFT, 450 small	Use high-contrast, glare-resistant display
Control Layout	Scattered, clustered, awkward	Ergonomic grouping by control priority
Navigation System	Linear, nested, joystick low	Introduce intuitive map hierarchy and
Visual Hierarchy	Flat layout, moderate layering	Multi-level hierarchy with contextual
Feedback System	Weak tactile, visual cues delayed	Integrate haptic + on-screen confirmation
Error Recovery	Minimal feedback, unclear prompts	Unified system error feedback layer
Consistency	Icons differ across bikes	Universal adventure-bike HMI icon set
User Freedom	Deep menus, joystick limits	Quick-access wheel or mode shortcut
Information Density	Sparse (NX), balanced (KTM), cluttered (RE)	Moderate density with adaptive data
Aesthetic Balance	Minimal vs. Overcrowded	Minimalist layout with contextual detail

Cumulative Evaluation Summary
Result Interpretation:
Your combined heuristic evaluation led to an integrated HMI recommendation for future adventure motorcycles:
"A visually hierarchical, ergonomically optimized, and feedback-rich interface system that supports rider intuition during dynamic touring conditions."
This synthesis ensures that future HMI designs prioritize ease of interaction, safety feedback, and rider situational awareness.

Individual Findings and Insights
(Cumulative of of 4 Personas)

1. Honda NX200 (2025)
Problems:
Limited feature set for adventure touring context.
Small display readability under direct sunlight.
Buttons for ABS and traction control not intuitive.
Cluster lacks hierarchical arrangement of information.
Minimal feedback when toggling settings.

Recommendations & Insights:
Introduce diversified control layout suited for gloves.
Improve display brightness and contrast ratio.
Provide clear tactile feedback on critical buttons.
Include feature hierarchy for quick access during rides.
Integrate trip data and riding mode icons visibly.

2. KTM Adventure 390S (2025)
Problems:
ABS settings buried deep within sub-menus.
Cruise control switch placement near throttle causing confusion.
Overloaded interface with too many layers of navigation.
Hierarchy in TFT layout not intuitive.
Key riding data not prioritized visually.

Recommendations & Insights:
Simplify navigation hierarchy; limit sub-menus.
Relocate cruise control switch ergonomically.
Enable quick toggles for ABS & traction.
Highlight riding mode indicators centrally.
Integrate adaptive brightness for better usability in daylight.

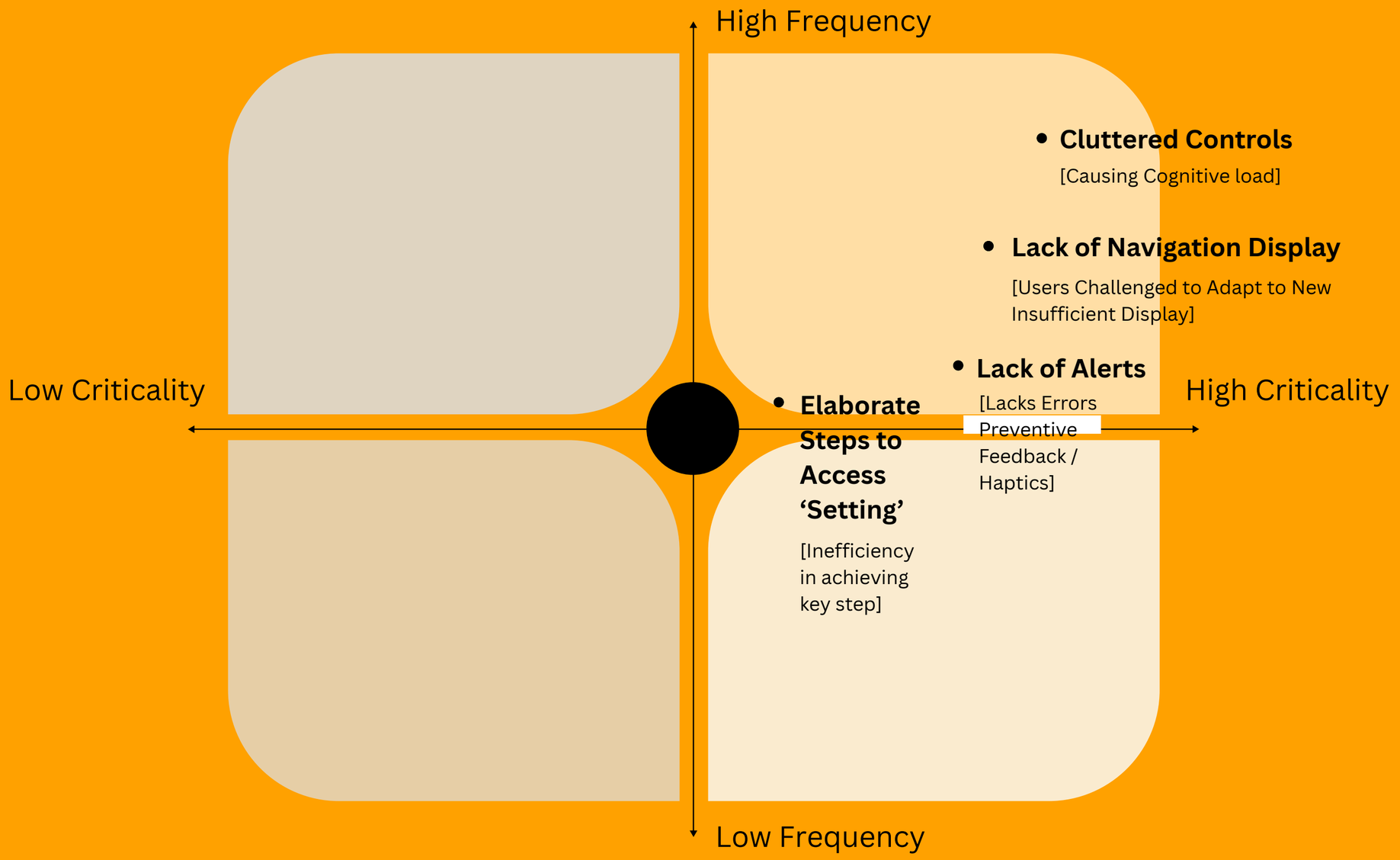
3. Royal Enfield Himalayan 450 (2025)
Problems:
Small icons and font size in display interface.
Navigation joystick positioned too low.
Hazard switch on opposite side of console.
Lack of visual hierarchy on main screen.
Limited feedback for interaction errors.

Recommendations & Insights:
Relocate joystick closer to thumb reach.
Increase display size and font contrast.
Align button positions according to ergonomic zones.
Introduce visual depth cues to improve hierarchy.
Provide audio or haptic feedback on input confirmation.

Cumulative table of individual assesment of Insights and recommendation

Category	Common Problems Identified	Consolidated Recommendations / Insights
Display Visibility	Poor readability in sunlight, small icons	Use anti-glare TFT with adaptive brightness
Button Placement	Critical controls not ergonomically grouped	Cluster buttons by frequency of use
Hierarchy of Information	Lack of intuitive screen structure	Redesign UI with layered visual hierarchy
Navigation Interaction	Deep menu nesting	Introduce quick-access shortcuts
Feedback System	Lack of tactile or visual confirmation	Add haptic or LED feedback on interaction
Error Prevention	Accidental switch activation	Separate safety controls spatially
Consistency	Inconsistent symbol usage across bikes	Standardize icon language
User Control	Riders lack direct control over modes	Allow easy access to key settings mid-ride
Cluster Aesthetics	Overcrowded information layout	Adopt minimalist visual design
Information Prioritization	Secondary data overpowering critical info	Focus main screen on speed, gear, and navigation

Problem Prioritization - Frequency vs Criticality



Problem–Impact–Feasibility Prioritization Matrix
(Based on Nielsen's Heuristic Principles applied to NX200, KTM ADV 390S, and Himalayan 450)

Problem / Issue	Associated Heuristic	User Impact (1–5)	Fix Feasibility (1–5)	Priority Level	Linked Insight
Poor visibility in sunlight (NX200, Himalayan)	Visibility of System Status	5	4	🔴 High	Upgrade TFT with adaptive brightness
Deep menu navigation for ABS & traction (KTM,	User Control & Freedom	5	3	🔴 High	Add quick-access shortcuts
Small or unclear icons (RE, NX200)	Recognition Rather Than Recall	4	4	🟠 Medium–High	Standardize iconography
Inconsistent symbol design (RE)	Consistency & Standards	4	4	🟠 Medium–High	Create unified symbol library
Low joystick position (Himalayan)	Ergonomic Accessibility	4	3	🟠 Medium–High	Relocate joystick within thumb reach
Overloaded display (RE)	Aesthetic & Minimalist Design	3	3	🟡 Medium	Simplify interface through layering
Lack of tactile feedback (NX200)	Error Prevention	3	4	🟡 Medium	Introduce haptic cues
Missing real-time help (All)	Help & Documentation	2	5	🟢 Low	Add digital help overlay
Cruise control near throttle (KTM)	Error Prevention	4	2	🟠 Medium–High	Reposition switch ergonomically
Lack of hierarchical organization (NX200, RE)	Match System & Real World	5	3	🔴 High	Redesign info hierarchy
Minimal feedback for system actions (All)	Visibility of System Status	4	4	🟠 Medium–High	Add LED/haptic feedback confirmation
Poor consistency in mode icons (All)	Consistency & Standards	4	4	🟠 Medium–High	Universal icon and color scheme
Lack of quick recovery from input errors (All)	Error Recovery	3	3	🟡 Medium	Contextual undo option
Visual clutter in RE cluster	Aesthetic & Minimalist Design	3	3	🟡 Medium	Reduce density, improve grouping
Weak night-time readability (NX200)	Visibility of System Status	3	4	🟡 Medium	Adaptive backlight modes

Priority Legend:
- 🔴 **High Priority: Critical usability or safety issue affecting rider interaction**
- 🟠 **Medium–High: Noticeable discomfort or confusion under real-world use**
- 🟡 **Medium: Minor inconvenience with improvement potential**
- 🟢 **Low: Optional enhancement or aesthetic refinement**

Linking Map — From Individual Problems to Combined Solution

Link Chain	Individual Issue	Heuristic Principle	Combined System Recommendation
Display → Readability	NX200 dim + Himalayan small icons	Visibility of System Status	Unified anti-glare TFT with adaptive brightness
Controls → Depth of Menus	KTM deep ABS menu + RE joystick low	User Control & Freedom	Add quick toggles and ergonomic joystick
Symbol → Recognition	RE unclear icons + NX200 minimal UI	Recognition Rather Than Recall	Standardize symbols across all models
Layout → Consistency	RE inconsistent, KTM clear	Consistency & Standards	Establish brand-independent iconography guide
Feedback → Responsiveness	NX200 no feedback + RE delayed	Error Prevention	Implement haptic + LED feedback
Density → Clarity	RE overloaded + NX200 simple	Aesthetic & Minimalist Design	Adaptive info density scaling
Errors → Recovery	All limited feedback	Help Users Recover from Errors	Add real-time correction prompts
Ergonomics → Accessibility	Himalayan joystick + KTM throttle	Match Between System & Real World	Place critical controls within thumb reach
Mode Awareness	KTM clear + RE unclear	Visibility of System Status	Consistent visual indicator for riding modes
🔟 System Learning	All limited help	Help & Documentation	On-screen help & quick tutorial overlay

- **Top 5 critical improvements that emerged are:**
- Adaptive high-contrast display system for all environments.
- Ergonomically placed quick-access controls for ABS, traction, and navigation.
- Standardized iconography and feedback patterns across all bikes.
- Multi-level interface hierarchy for intuitive navigation.
- Integrated feedback and recovery system ensuring safety and awareness.

Problem Goal Mapping

Problem	Evidence (from Part I)	Impact	Design Goal	Success Criteria
1. Cluttered Controls (Causing Cognitive Load)	Heuristic Evaluation – Violation of Heuristic #8: Aesthetic and Minimalist Design; Cognitive Walkthrough – Users hesitated identifying correct control while riding.	**High** – Increased mental effort, distraction while riding, delayed reaction times.	Simplify control layout by grouping related functions and eliminating redundant switches; ensure logical placement and labeling.	Users identify and operate correct control within 2 seconds without visual search; reduced reported mental load in walkthrough.
2. Lack of Navigation Display (Users Challenged to Adapt to New Insufficient Display)	Comparative Evaluation & User Feedback – Users reported difficulty understanding route info; Heuristic #6 (Match between system and real world).	**High** – Users cannot rely on system for guidance, leading to safety issues and dissatisfaction.	Integrate intuitive navigation visuals (map or directional cues) that resemble familiar smartphone layouts; maintain readability under sunlight.	Riders can interpret navigation cues accurately during test ride without external aid; navigation feature rated “easy to use” by >80% users.
3. Lack of Alerts (Lacks Error Preventive Feedback / Haptics)	Heuristic #5: Error Prevention & Feedback; Cognitive Walkthrough – No indication of critical events (e.g., ABS status, side stand, service alerts).	**Medium–High** – Missing feedback may cause missed warnings and unsafe conditions.	Incorporate multimodal feedback (visual + auditory + optional haptic) for critical alerts and confirmation prompts.	Riders acknowledge alerts instantly in simulation; no missed warnings during evaluation; reduced error rate by >50%.
4. Elaborate Steps to Access ‘Settings’ (Menu Inefficiency)	Heuristic #7: Flexibility and Efficiency of Use; Task Analysis – Users required >4 steps to reach basic settings.	**Medium** – Increases frustration, distracts from primary task (riding).	Redesign menu structure with shallow hierarchy; add shortcut or quick-access control for frequently used settings.	Access time reduced by 50%; users reach settings within 2 clicks or <3 seconds during usability test.

Display and Control Design Principles

Problem	Relevant Principles	How Principles Address the Issue / Improve User Experience
1. Cluttered Controls (Cognitive Load)	Cognitive Load Reduction, Visual Hierarchy, Clarity, Consistency, Efficiency	Simplifies layout by grouping related elements and maintaining consistent structure, reducing mental effort and distraction. Improves visibility, quick control recognition, and overall focus while riding.
2. Lack of Navigation Display	Clarity, Feedback, Contextual Relevance, Contrast, Predictive Aiding	Introduces clear, high-contrast navigation visuals with real-time cues placed contextually near key ride info. Enhances route understanding, safety, and ease of adaptation.
3. Lack of Alerts / Haptic Feedback	Feedback, Status, Redundancy Gain, Error Prevention, Accessibility	Adds visual, audio, and haptic alerts for critical events. Improves awareness, reduces missed warnings, and ensures all users receive timely feedback.
4. Complex Steps to Access Settings	Cognitive Load Reduction, Consistency, Contextual Relevance, Efficiency, User Control	Reduces menu layers and groups options logically, adding shortcuts for frequent tasks. Enables faster, more intuitive interaction and greater user satisfaction.

Control / Input	Current Effect on Display / System	Mapped Issue (from Part I)	Proposed Improvement / Enhancement
Mode / Select Button	Cycles through trip info, fuel data, clock sequentially	*Elaborate steps to access settings; high cognitive load*	Introduce a quick-access or long-press shortcut for “Settings.” Group related options under submenus for faster access.
Navigation / Info Button	Scrolls through data screens repeatedly	*Cluttered controls; inefficient info browsing*	Merge related functions (Trip/Fuel/ODO) under one control; add on-screen indicator for active mode to reduce confusion.
Indicator & Horn Cluster	Standard functions but closely spaced	*Cluttered control layout*	Increase spacing and tactile differentiation between switches for safer, eyes-free operation.
ABS / Warning Icons	Passive display indicators only	*Lack of alerts or feedback*	Integrate mild audio or haptic feedback for safety alerts; use brighter, larger symbols with confirmation cues.
Navigation Integration (via app)	Limited or absent on display	*Lack of navigation display / feedback*	Add minimal, context-aware navigation cues on screen with handlebar-based route interaction (pause, zoom, next-turn).

How Might We

HMW Statements

Less Cluttered Controls
How might we simplify and reorganize the switchgear layout for riders navigating complex control options in order to reduce confusion and cognitive load during rides?

Giving More Noticeable Cues for the Rider
How might we enhance visual and haptic feedback cues for riders performing key actions like mode changes or warnings in order to ensure immediate recognition and safer response times?

Fewer Steps to Achieve a Single Task
How might we streamline menu navigation and control sequences for riders interacting with settings or riding modes in order to minimize the number of steps and distraction while riding?

Fewer Steps to Achieve a Single Task
How might we make the interface quicker and easier to use for riders in order to reduce the steps needed to complete a task?

Displaying Navigation in a More Conveying and User-Friendly Way
How might we design a more informative and user-friendly full navigation display for riders in order to make route information clearer and easier to follow while riding?

VALIDATION

Nielsen’s Heuristic	Evaluation Severity (0–4)	Example / Observation
1. Visibility of system status	0	Real-time Google Maps, clear indicators. Big enough screen.Good contrast. icons are well lit and visible in sunlight.
2. Match between system and real world	0	Natural and familiar interface. No complex Icons or terms are used.
3. User control and freedom	1	Easy to switch Eco/Sport or ABS on/off.
4. Consistency and standards	1	Some controls differ slightly from global standards, but are easier and learnable.
5. Error prevention	0	Enough breathing space is given between controls.
6. Recognition rather than recall	0	Riders see the map directly instead of remembering arrows.
7. Flexibility and efficiency of use	1	Easy accessible controls and faster task achievement.
8. Aesthetic and minimalist design	0	Multi-functional buttons are there. Avoids cluttered controls.
9. Help users recognize, diagnose, recover	1	Essential warnings and updates are there.They are conveyed well to the rider through visual and haptic cues.
10. Help and documentation	1	Guidance for usage of multifunctional buttons are displayed on screen without leaving the rider confused.